MILITARY SPECIFICATION

CORRECTOR, CANT: M4E1

1. SCOPE

1.1 This specification covers one type of instrument known as the M4E1 Cant Corrector. This instrument is equipped with a level vial which is radioactively illuminated.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

F11730986	Corrector, Cant: M4E1

PACKAGING DATA SHEET

11730986	Packaging of Corrector, Cant: M4E1

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:23Z
Check the source to verify that this is the current version before use.

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for porposal shall apply.

DEFENSE SUPPLY AGENCY MANUAL

DSAM 4145.8 Radioactive Commodities in the DOD Supply Systems

(Copies of DSAM 4145.8 may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Regulations pertaining to radioactive devices.- Marking, identification, handling, and shipping of radioactive devices shall be in accordance with DSAM 4145.8.

3.2 Fabrication.- The cant corrector shall be manufactured in accordance with Drawing F11730986.

3.3 General specification.- The following requirements of MIL-F-13926 shall be met:

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.4 Environmental.

3.4.1 Storage temperatures.- The cant corrector shall show no evidence of physical failure due to being thermally stabilized at -80° and +160° Fahrenheit (F).

3.4.2 Operating temperatures.- The cant corrector shall meet the applicable requirements of 3.5.2 while thermally stabilized at -40° and +150°F.

3.4.3 Vibration.- With the cant corrector locked against rotation, the cant corrector shall be vibrated at a constant frequency of thirty cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a periof of 5 minutes ± 15 seconds. Subsequent to vibration, the lock shall have remained engaged and there shall be no evidence of damage or physical failure.

3.5 Orientation.- The cant corrector shall be mounted on a horizontal surface, the crosslevel vial centered and the plane of the gunner's quadrant pad horizontal within 0.10 mil.

Source: https://assist.dla.mil - Downloaded: 2026-10-22T20:23Z
Check the source to verify that this is the current version before use.

3.5.1 Operability.- With the cant corrector thermally stabilized at a temperature between 70° and 75°F, the torque required to rotate Housing 8626160 shall be between 8 and 16 ounce-inches. When exposed to the operating temperatures specified in 3.4.2, the torque required to rotate Housing 11731017 shall not exceed 5 inch-pounds.

3.5.2 Excursion.- The housing section containing the level vial shall move through an excursion of at least 230 mils in one direction and 1383 mils in the opposite direction from the position established in 3.5 when the rotation lock is disengaged.

3.5.3 Lock.- The rotational lock, when engaged, shall prevent rotation of the housing section.

3.6 Level vial.- The level vial bubble shall be visible in all ambient light conditions.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwide specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwide specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) cant correctors plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:23Z
Check the source to verify that this is the current version before use.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL), when provided with the technical data package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Table II herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Table II shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

Class	Characteristic	Requirements	Test Procedure
Critical:	None		
Major:	1.0% Defective		
101.	Vibration	3.4.3	4.6.2, 4.6.2.1
102.	Operability	3.5.1	4.6.6
103.	Excursion	3.5.2	4.6.4
104.	Lock	3.5.3	4.6.5
105.	Level vial illumination	3.6	4.6.7
Minor:	None defined		

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:23Z
Check the source to verify that this is the current version before use.

and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- Samples shall be selected at random from lots listed using inspection level S-3 Table I of MIL-STD-105. Acceptance or rejection of the lot shall be in accordance with AQL 2.5 of Table II-A of MIL-STD-105. The samples shall have met all the requirements and tests outlined in Table I prior to being subjected to the following examination and test. The requirement outlined in 3.1 and 3.2 shall be accepted on contractors certification.

TABLE II

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Regulation pertaining to radioactive materials	3.1	Certification
302. Fabrication	3.2	Certification
303. General specification	3.3	Certification
304. Temperature storage	3.4.1	4.6.1.1
305. Temperature operating	3.4.2	4.6.1.1
306. Operability (Torque at oper temp)	3.5.1	4.6.6

4.4.2 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the Technical Data Package (TDP).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:23Z
Check the source to verify that this is the current version before use.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerance used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment.- In conjunction with 4.5.2.2, the following standard test equipment shall be utilized in the performance of the applicable tests as specified in 4.6.

NOMENCLATURE	DESCRIPTION
1. Hot and Cold Chamber	Standard type conforming to the accuracies outlined in Test Facilities specified in MIL-F-13926.
2. Torque Medium	Standard shop type accurate to within 10% of Drawing Tolerance

4.6 Test Method and Procedure

4.6.1 Environmental

4.6.1.1 Temperature test.- The cant corrector shall be subjected to Test Procedure I of MIL-F-13926; however, the temperature specified in 3.4.1 and 3.4.2 shall apply. When the cant corrector has been stabilized at the temperature specified the result of each test shall conform to the limits outlined in 3.4.1 and 3.4.2 using the test procedure outlined herein.

4.6.2 Performance test.

4.6.2.1 Vibration.- Standard vibration equipment capable of providing the amplitude and frequency while vibrating in a vertical plane shall be utilized in conformance with the conditions specified in 3.4.3. Unless otherwise specified, the test equipment shall meet the "Test Facilities" requirements of Specification MIL-F-13926. The mounting surface of the corrector shall be in a horizontal plane within two (2) degrees when secured to the vibrating equipment and the locking mechanism of the corrector in the locked position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:23Z
Check the source to verify that this is the current version before use.

4.6.2.1.1 Post vibration.- After removing the cant corrector from the vibration equipment, the cant corrector shall be examined for loose, damaged or missing parts prior to being subjected to the following test. A tactile examination shall be made to determine excessive looseness or improper function of such moving parts as rotation of the member supporting the level vial when the assembly is unlocked, the level vial cover, and the push action of the knobs to lock and unlock the assembly, as well as screws, retainers, and other assembled parts which may have loosened as a result of vibration. Any one of the possible defects defined above shall be cause to consider a corrector as defective.

4.6.3 Orientation.- The cant corrector shall be supported in an appropriate holding adapter, secured to an elevation type test fixture capable of providing elevation in excess of 1400 mils in one direction and 250 mils in the opposite direction and shall provide elevation readings within an accuracy specified in 4.5.2.2. Prior to performing the test, level and cross-level, the fixture using a fire second precision level. Secure holding adapter and recheck level position of adapter seating surface using a five second level. Mount cant corrector with the locking mechanism in the unlocked position to the adapter. Set level vial of cant corrector to indicate level within the line thickness of the level vial. Place Inspection Aid A8658940 over the quadrant pads and determine levelness of pads with respect to the cant correctors seating surface, they shall be level within the limits outlined in 3.5. Measure error,if any,by means of the inspection aid.

4.6.3 Excursion.- Rotate level vial assembly to maximum position in one direction and then rotate the elevation fixture in the opposite direction to reposition the level vial bubble level. Repeat this procedure in the opposite direction. The readings at each limit shall be equal to or exceed the minimum readings outlined in 3.5.2.

4.6.5 Lock.- Using standard shop type tactile inspection techniques determine that the locking function of the cant corrector functions within the limits outlined in 3.5.3.

4.6.6 Operability.- Using standard shop type torque adapters and measuring media, determine that the torque required to rotate the housing specified in 3.5.1 is within the limit specified in 3.5.1 for the temperature specified.

4.6.7 Level vial .- Place the instrument in a dark room having a control light source. The observer and instrument must become dark adapted. After being adapted to the darkness the observer shall visually examine the level vial and graduation, they must be clearly discernable to assure conformance with 3.6. Gradually raise the level of light in the darkened room one (1) foot candle of illumination in the area of the level vial. The level vial and graduation shall be clearly discernable to assure conformance with 3.6.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 11730986. The level of protection shall be as specified in the procurement document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:23Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use.- The M4E1 Cant Corrector is primarily intended for mounting on the gun carriage of the 105MM Howitzer XM102 to provide a surface that is horizontal in the direction of the line of fire thus providing a mounting surface for a gunner's quadrant to check the angle of cant of the gun trunnion regardless of the elevation or depression of the gun tube without introducing level vial roll into the quadrant.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing (see 5.1).
(c) Applicable packaging data sheet number (see 5.1).
(d) Applicable stock number.

6.3 Radioactive material.- Prospective bidders should be cognizant of the need for an Atomic Energy Commission license governing the procurement and handling of radioactive material.

Custodian:

Army - MU

Preparing activy

Army - MU

Project No. 1290-A257

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:23Z
Check the source to verify that this is the current version before use.